

Payload Analysis NSE Main Website

Note: Payload's Response filters at 1 Year timeline, typically shows around 3-4 months data so it will be strongly recommended to put only 1 month timeline in the payload for accurate and precise response.

HISTORICAL INDEX DATA

<https://www.nseindia.com/api/historicalOR/indicesHistory> (Request URL, GET)

- NIFTY 50 - indexType=NIFTY%2050&from=29-03-2025&to=29-03-2026
- NIFTY BANK - indexType=NIFTY%20BANK&from=29-03-2025&to=29-03-2026
- NIFTY500 LARGEMIDSMALL EQUAL-CAP WEIGHTED - indexType=NIFTY500%20LARGEMIDSMALL%20EQUAL-CAP%20WEIGHTED&from=29-03-2025&to=29-03-2026

HISTORICAL INDEX YIELD

<https://www.nseindia.com/api/historicalOR/indicesYield> (Request URL, GET)

- NIFTY MIDCAP 100 - indexType=NIFTY%20MIDCAP%20100&from=01-03-2026&to=29-03-2026
- NIFTY500 HEALTHCARE - indexType=NIFTY500%20HEALTHCARE&from=29-12-2025&to=29-03-2026
- NIFTY50 DIVIDEND POINTS - indexType=NIFTY50%20DIVIDEND%20POINTS&from=29-09-2025&to=29-03-2026

Important Observations:

1. Historical Index Data - Each word is separated by a "%20" sign then "&from='start_date'&to='end_date'"
2. Historical Index Data's Payload format for any index is similar to Historical Index Yield's Data Payload format.

3. Only difference is the headers of both - which is around 80% same, just write - 'indicesHistory' or 'indicesYield'.
4. The Payload is only dependent on the Original Full Index Name and not on some other metrics like trading_index_name, etc resulting in more robust architecture and pipeline when built.

Deletion (ONLY NIFTYINDICES.COM):

1. NIFTY MIDSMALLCAP400 50:50 - Broad Market Indices - 13 - USED - ADDED
2. NIFTY SMALLCAP 500 - Broad Market Indices - 18 -USED - ADDED
3. NIFTY CEMENT - Sectoral Market Indices - 24 - USED - ADDED
4. NIFTY REITS & REALTY - Sectoral Market Indices - 43 - USED - ADDED
5. NIFTY INDIA CORPORATE GROUP INDEX - TATA GROUP - Thematic Market Indices - 101 - USED - ADDED - **NOT SHOWN ON NSE INDIA MAIN WEBSITE IN DROPDOWN BUT RETURNS DATA WHEN API IS HIT PROPERLY!!!**
6. NIFTY100 ESG SECTOR LEADERS - Thematic Market Indices - 129 - USED - ADDED

Addition (ONLY NSE MAIN WEBSITE):

1. NIFTY 1D RATE INDEX - Strategy Market Indices - 13 - ADDED

ONLY NSE MAIN WEBSITE:

1. NIFTY500 HEALTHCARE - Sectoral Market Indices - 44 - ADDED

Important Observation:

```
payload = {"cinfo":{"name":"NIFTY HEALTHCARE','startDate':'11-Mar-2026','endDate':'11-Mar-2026','indexName':'NIFTY HEALTHCARE'}}"
```

This is the original payload of niftyindices website to return data for that particular index

BUT

I did some sneaky thing and changed it slightly to

```
payload = {"cinfo":{"name":"NIFTY HEALTHCARE','startDate':'11-Mar-2026','endDate':'11-Mar-2026','indexName':'NIFTY HEALTHCARE INDEX'}}"
```

and this was still returning the data

that's why now in the table index_metadata, at index_id 31, index_long_name = 'NIFTY HEALTHCARE INDEX' and trading_index_name = 'NIFTY HEALTHCARE'

Since for NSE Main Website API attempt, we can just use the index_long_name and that will perfectly return the data and for Nifty Indices Website API attempt, we can use the modified trading_index_name then it will return the same data too!

Technical Audit: NiftyIndices API Reverse Engineering

Date: March 31, 2026

Subject: Payload Optimization & Input Validation Discovery

1. The Problem: "Silent" Data Gaps

While fetching historical data for the `NIFTY500 HEALTHCARE` index, the official exchange (niftyindices.com) frontend was returning **Null/Empty** responses. Upon intercepting the POST request, I identified that the frontend was sending a truncated string:

- **Failed Payload:** `{'name': 'NIFTY500 HEALTH', ...}`
- **Status:** Silent Failure (200 OK, but 0 rows returned).

2. The Discovery (Fuzzing & Testing)

I performed a series of "Manual Interceptions" to map the backend's validation logic. By modifying the `cinfo` string in the POST payload, I identified the **Minimum Viable Payload (MVP)** required to trigger a valid response in the following part of my Python Code (especially in the payload part) :

```
import requests

url = "https://www.niftyindices.com/Backpage.aspx/getHistoric
aldataabletoString"

payload = {"cinfo": "{ 'name': 'NIFTY500 HEALTHCARE', 'startDat
e': '07-Jan-2026', 'endDate': '11-Mar-2026', 'indexName': '' }" }
```

```

headers = {

    "Referer": "https://www.niftyindices.com/",

    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"

}

response = requests.post(url, json=payload, headers=headers)

data = response.json()

if response.status_code == 200:
    print("Intercepted", len(data), "rows of data.")
    print(data)
else:
    print("No Response Intercepted!")

```

Test Case	Payload Modification	Result	Engineering Insight
Full Name	<code>name: 'NIFTY500 HEALTHCARE'</code>	SUCCESS	The backend requires the Legal Long Name , ignoring the frontend's truncated version.
Fuzzy Match	<code>name: '... HEALTHCARE INDEX'</code>	SUCCESS	The backend uses a "Fuzzy" or LIKE match logic; trailing characters don't break the query.
Dummy Key	<code>indexName: 'hg'</code>	SUCCESS	The <code>indexName</code> parameter is Vestigial ; its value is ignored by the database.
Null Key	<code>indexName: ''</code>	SUCCESS	The key must exist, but the value can be empty.
Missing Key	<i>Deleted</i> <code>indexName</code> <i>key</i>	FAILURE	The server-side parser crashes if the key is missing (Likely a

			KeyError).
--	--	--	-------------

4. Final Conclusion

By "Red Teaming" the official API, I identified that the system relies on a **Single-Key Dependency** (`name`).

URL SYSTEM

Nifty Indices

1. Historical OHLC Data:

<https://www.niftyindices.com/Backpage.aspx/getHistoricaldatatabletoString>

2. PE, PB, Div. Yield Data:

<https://www.niftyindices.com/Backpage.aspx/getpepbHistoricaldataDBtoString>

NSE India

1. Historical Index Data:

https://www.nseindia.com/api/historicalOR/indicesHistory?indexType=NIFTY_50&from=19-10-2025&to=19-04-2026

2. Historical Index Yield: https://www.nseindia.com/api/historicalOR/indicesYield?indexType=NIFTY_50&from=19-04-2025&to=19-04-2026